

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) building developer-facing infrastructure for LLM agents — async job orchestration, multi-provider routing, and MCP servers. Designed SiliconCrew — a LangGraph agent with **async job orchestration**, multi-provider LLM routing (Gemini / OpenAI / Anthropic), and a full **MCP server (stdio / SSE / Streamable HTTP)** exposing tools to Claude Desktop, VS Code, and any MCP client. Solo-shipped Trezzit (trezzit.com), a production AI platform serving **600+ users** for 12+ months with 60+ REST endpoints, billing tiers, async pipelines, and per-provider cost / latency telemetry — live, on-call, on me.

TECHNICAL SKILLS

LLM Platform & Agents: LangGraph | ReAct Agents | LangChain | MCP Protocol (stdio / SSE / Streamable HTTP) | Multi-provider LLM Routing (Gemini / OpenAI / Anthropic) | Tool Use / Function Calling | Streaming APIs | RAG Pipelines | Pydantic Validation | LLM Evaluation & Observability

Backend, APIs & Job Orchestration: Python | FastAPI | Django REST Framework | Async Python (ThreadPoolExecutor) | WebSocket | PostgreSQL | Redis | Celery + Celery Beat | Per-run Isolation | Exponential Backoff | On-disk Run Recovery | Docker | DigitalOcean | Nginx | GitHub Actions CI/CD

Frontend & DevEx: React 19 | Next.js | TypeScript | Tailwind | Radix UI | Monaco Editor | Zustand | SDK / Developer Tooling | Linux | Git

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University

Tempe, AZ

Coursework: Artificial Intelligence · Reinforcement Learning in Robotics · Perception in Robotics · Foundations of Algorithms · Algorithm/Hardware Co-Design

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — Developer Platform for LLM Agents: Job Orchestration, MCP SDK & Streaming UI

Nov 2025 – Present

- Built a **full developer-facing platform** around a 21-tool LangGraph ReAct agent — FastAPI + WebSocket backend, multi-provider LLM routing (Gemini / OpenAI / Anthropic) with provider-specific quirk handling, full **MCP server** that auto-discovers LangChain @tool functions and exposes them over stdio / SSE / Streamable HTTP transports (compatible with Claude Desktop, VS Code, and any MCP client); public at github.com/naman-ranka/siliconcrew.
- Async job orchestration** for long-running LLM-driven workflows: ThreadPoolExecutor-based engine with per-run isolation, exponential backoff (30s→10min), 20-min hard timeout, on-disk index for process-restart recovery, and three sign-off guardrails before completion; autonomous self-correction loop (read diagnostic → patch → re-verify) achieves **37/92 first-pass on CVDP** (Comprehensive Verilog Design Problems) and **5/5 on ASU Spec2Tapeout**.
- Developer UI in **Next.js 14 + TypeScript + Zustand**: 3-panel client (Sidebar | Chat | Artifacts) with Monaco code editor, 6 artifact viewer tabs (spec, code, waveform, schematic, layout, report), WebSocket-streamed tool-call execution cards rendering live, and per-session token + cost tracking across all LLM providers.

Trezzit — Production AI Platform with Billing, API Surface & MCP Server, Sole Engineer (trezzit.com)

Feb 2025 – Present

- Solo-shipped Trezzit and run it on-call for **12+ months** — React 19 + Django platform serving **600+ users** with **60+ REST endpoints**, **40 DB models**, **995 tests**; provider-abstracted LLM registry across Gemini / OpenAI / Claude / Grok with a two-pass pipeline (extract → semantic validation → model-upgrade fallback) iterated from ~60% → **95%+** accuracy.
- Built an **MCP server** exposing the platform API to AI assistants (create bills, query balances, upload receipts) over the same auth and data layer as the web app; **billing-tier system** (Student free, Pro \$5/mo) with rate limits (100/hr anonymous, 10k/hr authenticated) and cookie JWT + OAuth + OTP auth.
- Production reliability + observability: GitHub Actions CI with live PostgreSQL service container, prod→staging DB sync, encrypted backups; Redis caching, Celery + Celery Beat for async background jobs; observability tracks per-endpoint p95 latency and per-provider tokens / cost across all 60+ endpoints.

HEAL.AI — Healthcare AI with Custom RAG Pipeline

Sep – Nov 2025

- Won 1st place at Devlabs Hackathon; built a custom RAG pipeline without an external vector database — PDF chunking with 2-sentence overlap, Google text-embedding-004 (768-dim) embeddings, cosine similarity retrieval with confidence scoring, and automated FDCPA-compliant dispute email generation via Gemini; FastAPI backend, React + TypeScript frontend.

WORK EXPERIENCE

Software Engineer (Sole Engineer)

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won **‘Best Use of AI’** at Strategy X DevHacks’25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI healthcare assistant with a custom RAG pipeline for insurance Q&A and automated medical bill dispute generation.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for password generation.